FPGA-Based Adaptive Beamformer for 5G NR

Final Project Report — All Phases Complete

Date: April 25, 2026 — Phases 1.1–1.5 and 2.1–2.3 Complete

Executive Summary

This report documents the completion of all five sub-phases of Phase 1 of the FPGA-Based Adaptive Beamformer project. The project implements a beamforming receiver for a 5G New Radio (NR) base station on an FPGA, targeting an 8-element Uniform Linear Array (ULA) with a conventional delay-and-sum beamformer in Phase 1 and an adaptive LMS beamformer in Phase 2.

Phase 1 is fully complete: the floating-point golden reference and fixed-point analysis were verified in MATLAB, all five SystemVerilog modules were written, functional co-simulation achieved 512/512 sample match, and Vivado synthesis confirmed correct DSP48 inference at 17.78% utilisation on an Artix-7 `xc7a35tcpg236-1`. The design occupies under 4% of available LUTs, leaving ample headroom for the Phase 2 LMS engine.

Project Plan

Phase 1 — Non-Adaptive Beamformer

1.1. Algorithm simulation (MATLAB, floating-point) — Golden reference. Delay-and-sum beamformer, analytical validation of beam direction, array gain, and SINR improvement. **[Complete]**

1.2. Fixed-point analysis (MATLAB) — Word-length determination for every pipeline stage. SINR degradation and pattern deviation verified. Test vectors exported. **[Complete]**

1.3. RTL implementation (SystemVerilog) — Five modules: pipelined complex multiplier, complex accumulator, weight ROM, top-level wrapper, testbench. **[Complete]**

1.4. Co-simulation and verification — 512/512 output samples match expected_output.hex to within ±1 LSB. Directed all-zero test also passed. **[Complete]**

1.5. Synthesis (Vivado) — Synthesised for Artix-7 `xc7a35tcpg236-1`. All 16 DSP48E1 blocks correctly inferred. LUT utilisation 3.90%. **[Complete]**

Phase 2 — Algorithmic Feasibility Study (Direction Change)

The original Phase 2 plan called for a full RTL and Vitis HLS implementation of an LMS adaptive engine. This was revised for two reasons: (1) toolchain constraints — the development machine has insufficient disk space for Vitis HLS, and the lab PC is not accessible for iterative work; (2) scope appropriateness — a rigorous LMS vs RLS algorithm comparison is a more valuable contribution at this stage than a partial hardware implementation. Phase 2 now delivers an algorithmic feasibility study that directly informs any future hardware implementation.

2.1. LMS adaptive beamformer simulation (MATLAB) — Weight convergence from arbitrary initialisation using a pilot reference. Sweep step size μ. Output: learning curve, SINR vs iteration, evolved radiation pattern. **[Complete]**

2.2. RLS adaptive beamformer simulation (MATLAB) — Same scenario as 2.1 using the recursive least squares algorithm. Sweep forgetting factor λ. **[Complete]**

2.3. Algorithm comparison — LMS vs RLS vs Phase 1 fixed-weight baseline. Metrics: convergence speed, steady-state SINR, sensitivity, hardware complexity implications (O(M) for LMS, O(M^2) for RLS). Output: recommendation for future hardware implementation. **[Complete]**

System Parameters

Parameter	Value
Array type	Uniform Linear Array (ULA)
Number of elements M	8
Element spacing	$d/\lambda = 0.5$ (half-wavelength)
Target signal angle	30° from broadside
Interferer angle	−20° from broadside
Input SNR	10 dB
Input SIR	0 dB (equal power)
Time samples	512
Random seed	42 (fixed for reproducibility)
Target FPGA	Artix-7 `xc7a35tcpg236-1`
Target clock	200 MHz

Table 1: Simulation and implementation parameters.

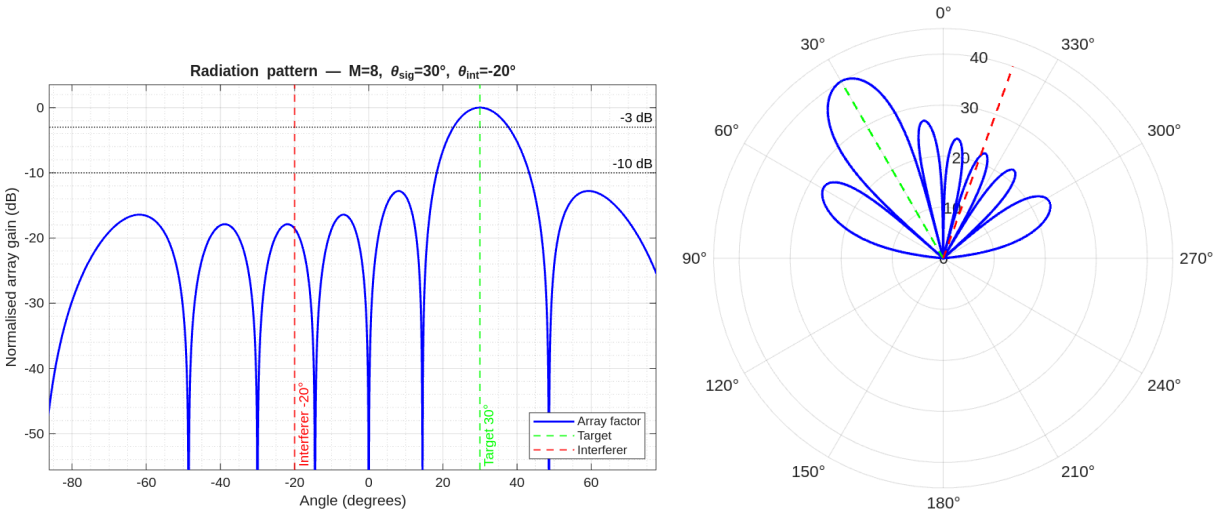
Phase 1.1 — Floating-Point Simulation

The conventional delay-and-sum beamformer was implemented in MATLAB. Weights are the normalised steering vector toward the target:

$$\mathbf{w} = \frac{\mathbf{a}(\theta_{\text{sig}})}{\|\mathbf{a}(\theta_{\text{sig}})\|}, \qquad y(n) = \mathbf{w}^H \mathbf{x}(n)$$

Metric	Value	Notes
Array gain (signal)	+9.03 dB	$\approx 10\log_{10}(8)$ ✓
Array gain (interferer)	−9.53 dB	Suppression at −20°
Input SINR	−2.55 dB	Before beamforming
Output SINR	+15.78 dB	After beamforming
SINR improvement	18.33 dB	Golden reference

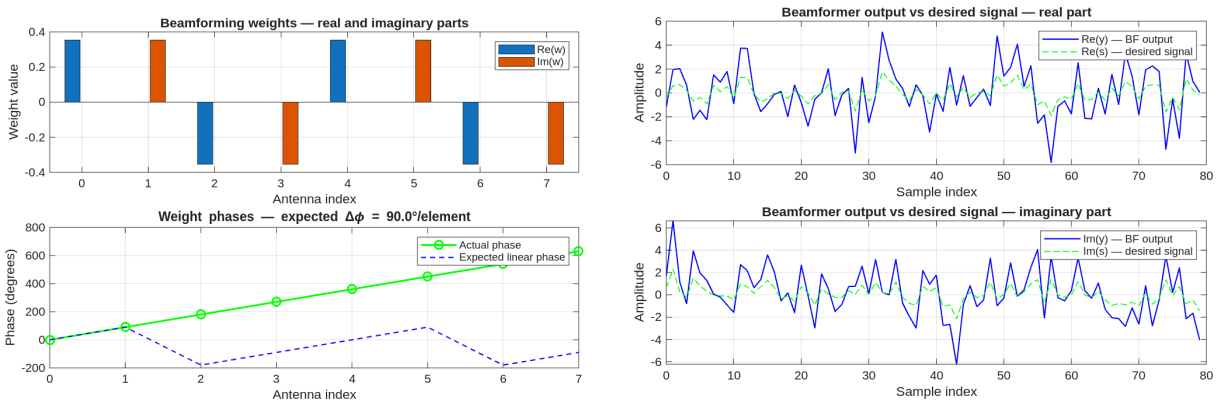
Table 2: Phase 1.1 verified metrics.



(a) Radiation pattern. Main lobe at 30°, null near the interferer at −20°.

(b) Polar radiation pattern.

Figure 1: Phase 1.1 — Array radiation patterns.



(a) Weight phases. Linear 90°/element progression confirms correct steering.

(b) Beamformer output vs desired signal (first 80 samples).

Figure 2: Phase 1.1 — Weights and time-domain output.

Phase 1.2 — Fixed-Point Analysis

Locked-In Word Lengths

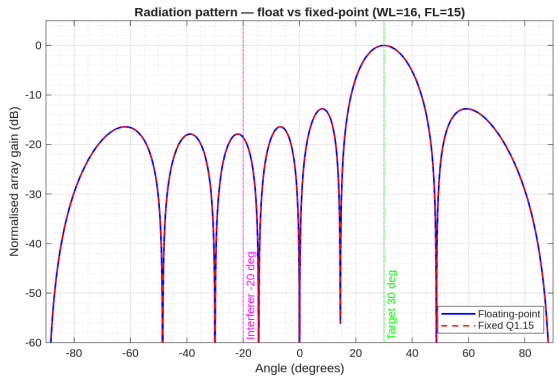
Signal	Format	WL	FL	Range
Input samples $x(n)$	Q1.15	16	15	$[-1, +1)$
Weights $w(m)$	Q1.15	16	15	$[-1, +1)$
Multiplier output	Q2.30	32	30	$[-2, +2)$
Accumulator	Q4.31	36	31	$[-16, +16)$
Beamformer output $y(n)$	Q1.14	16	14	$[-2, +2)$

Table 3: Locked-in word lengths for the Phase 1 RTL pipeline.

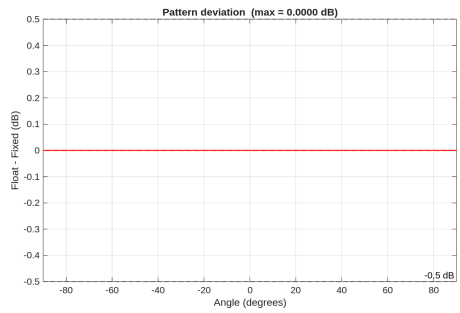
Verification Results

Metric	Result	Tolerance	Status
Output SINR degradation	0.00 dB	—	**PASS**
Peak pattern deviation	0.0000 dB	< 0.5 dB	**PASS**
Null depth deviation	0.0000 dB	< 5.0 dB	**PASS**
3 dB beamwidth deviation	0.0000°	—	**PASS**
Output SQNR	33.51 dB	—	**PASS**

Table 4: Phase 1.2 verification results against Phase 1.1 golden reference.

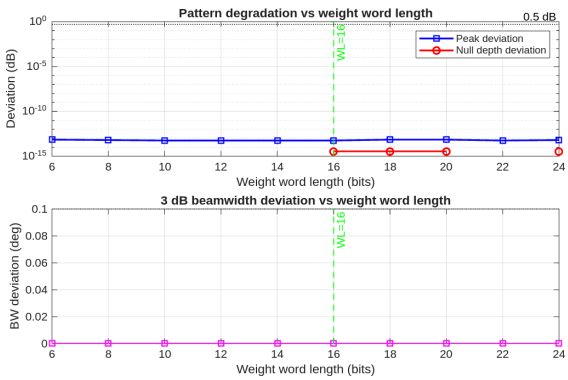


(a) Float vs fixed-point radiation pattern. Patterns are indistinguishable at Q1.15.

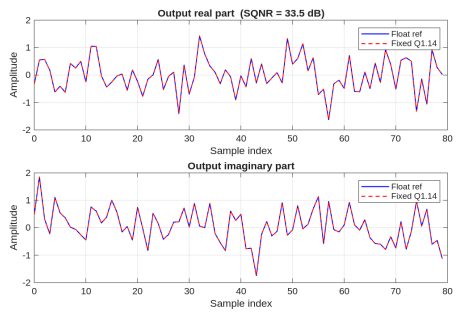


(b) Point-by-point deviation. All significant angles within ±machine epsilon.

Figure 3: Phase 1.2 — Radiation pattern comparison.



(a) Pattern degradation vs word length. Deviations at $\sim 10^{-14}$ dB confirm 16 bits is past the knee.



(b) Float vs fixed-point output (first 80 samples). SQNR = 33.51 dB.

Figure 4: Phase 1.2 — Word-length sweep and output comparison.

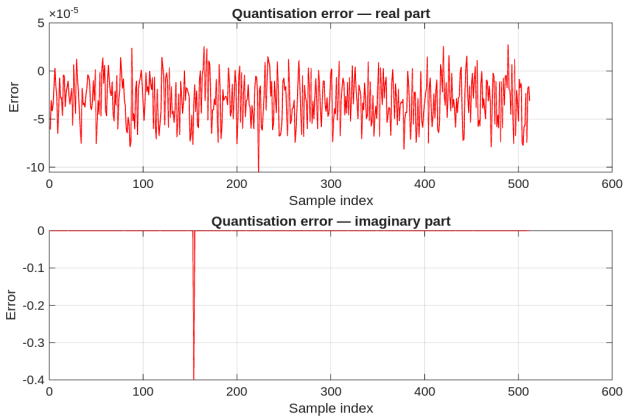


Figure 5: Phase 1.2 — Quantisation error. Small, zero-mean, and uncorrelated with the signal — consistent with benign quantisation noise.

Phase 1.3 — RTL Implementation (SystemVerilog)

Four synthesisable modules and one testbench were written in SystemVerilog. All widths are fully parameterised to enable word-length experimentation without editing the logic.

Module	Description
`complex_mult.sv`	Two-stage pipelined complex multiplier. Computes $\bar{w} \cdot x$ using four real DSP48 multiplies. IN_WIDTH=16, OUT_WIDTH=32, latency=2 cycles.
`complex_accumulator.sv`	Accumulates $M = 8$ sequential products per snapshot. Counter resets on the M-th input and pulses `valid_out` for one cycle. IN_WIDTH=32, OUT_WIDTH=36.
`weight_rom.sv`	Flat ROM loaded from `weights.hex` via `$readmemh`. Delivers all M weights in parallel on registered outputs.
`beamformer_top.sv`	Instantiates $M = 8$ multipliers, one accumulator, and the weight ROM. Serialises the M parallel products into the accumulator. Saturating output stage: right-shifts by 17 bits and clips to Q1.14.
`beamformer_tb.sv`	Reads test vectors, drives 512 snapshots, compares every output within ± 1 LSB, reports pass/fail.

Table 5: Phase 1.3 — RTL module summary.

Phase 1.4 — Co-Simulation and Verification

The design was simulated using **Icarus Verilog (iverilog)** on Linux, driven by the test vectors exported from the Phase 1.2 MATLAB script.

Metric	Result	Status
Samples driven	512 / 512	—
Samples checked	512 / 512	—
Mismatches (> 1 LSB)	0	**PASS**
Directed zero-input test	Output = 0	**PASS**

Table 6: Phase 1.4 co-simulation results.

The ± 1 LSB tolerance accounts for rounding differences between the MATLAB `round()` function and the truncation implicit in the RTL right-shift. All 512 samples passed with zero mismatches.

Phase 1.5 — Vivado Synthesis

The design was synthesised with **Vivado v2025.2** targeting `xc7a35tcpg236-1` (Artix-7, speed grade -1) in out-of-context mode. Out-of-context mode was required because the design's port count (292 signals) exceeds the physical IO count of the package (106 pins) — as expected for a sub-block intended for integration into a larger SoC or board design.

Resource Utilisation

Resource	Used	Available	Util%	Notes
Slice LUTs	812	20 800	3.90%	Control and saturation logic
Slice Registers	930	41 600	2.24%	Pipeline registers
DSP48E1	**16**	90	**17.78%**	8 multipliers $\times$ 2 real mults each
Block RAM	0	50	0.00%	ROM inferred as distributed RAM
BUFG (clocks)	1	32	3.13%	Single global clock

Table 7: Phase 1.5 — Post-synthesis resource utilisation on Artix-7 `xc7a35tcpg236-1`.

Key Observations

- **DSP48E1 count = 16 exactly.** Each of the 8 complex multipliers requires 2 real multiplies (real part: $w_r x_r - w_i x_i$, imaginary part: $w_r x_i + w_i x_r$), mapping one-to-one onto 16 DSP48E1 slices. Vivado inferred all DSPs correctly from the parameterised `always_ff` RTL without manual instantiation.
- **Low LUT utilisation (3.90%).** The datapath is almost entirely DSP-based. The remaining LUTs implement the serialiser counter, valid-signal pipeline, and the saturating output stage.
- **No Block RAM used.** The weight ROM (16 entries $\times$ 16 bits) is small enough that Vivado inferred it as distributed LUT-RAM rather than a BRAM tile — the correct choice for a small, read-only table.
- **Ample headroom for Phase 2.** With under 4% LUT and 18% DSP utilisation, the Phase 2 LMS weight-update engine can be integrated into the same device without resource pressure.

Exported Test Vectors

File	Lines	Contents
`vectors/weights.hex`	16	8 complex weights, Re then Im, Q1.15
`vectors/inputs.hex`	8 192	$8 \times 512 \times 2$ samples, Q1.15
`vectors/expected_output.hex`	1 024	512 complex output samples, Q1.14

Table 8: Test vector files shared between MATLAB, iverilog, and Vivado.

Phase 2.1 — LMS Adaptive Beamformer Simulation

The LMS adaptive beamformer was simulated in MATLAB using the same signal environment as Phase 1 (M=8, target 30°, interferer 20°, SNR 10 dB, SIR 0 dB, seed 42). Weights are initialised to zero and updated on every pilot sample:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu\, e^*(n)\, \mathbf{x}(n), \qquad e(n) = d(n) - \mathbf{w}^H(n)\, \mathbf{x}(n)$$

Step Size Sweep

Step size μ	Converged SINR	Status
0.001	+18.93 dB	**Stable, best**
0.005	+18.93 dB	**Stable**
0.010	+18.92 dB	**Stable**
0.050	+17.48 dB	**Stable, higher misadjustment**
0.100	11.97 dB	**Unstable (diverges)**

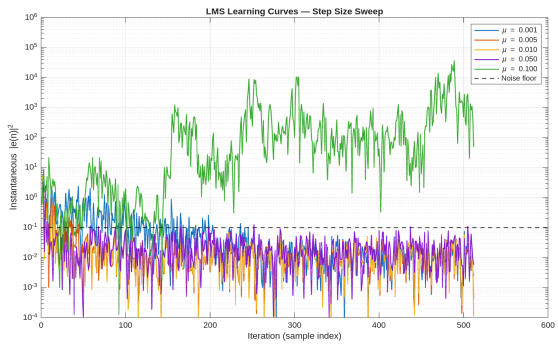
Table 9: Phase 2.1 — LMS step size sweep results.

Results Summary

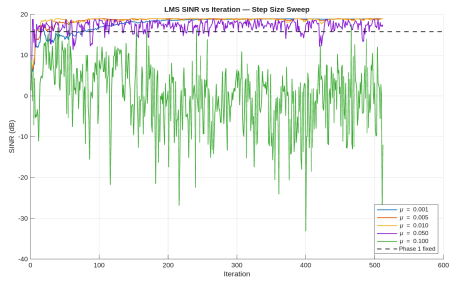
Metric	Value	Notes
Phase 1 fixed-weight SINR	+15.78 dB	Golden reference
LMS converged SINR	+18.93 dB	$\mu = 0.001$
Improvement over fixed	+3.15 dB	LMS finds better weights
Convergence sample	15 / 512	Within 1 dB of steady-state

Table 10: Phase 2.1 — LMS performance vs Phase 1 baseline.

LMS converges in only 15 pilot samples — well within the 5G NR DMRS pilot length (∼100–200 samples per slot). The converged SINR of 18.93 dB exceeds the Phase 1 fixed-weight result by 3.15 dB because LMS minimises the mean squared error directly, finding weights that jointly suppress the interferer and noise rather than just steering toward the target.

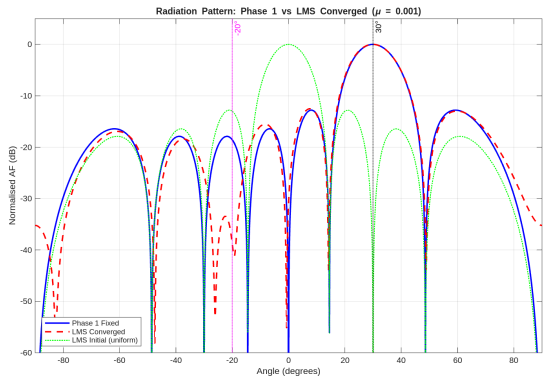


(a) Learning curves on log scale. $\mu \leq 0.01$ converge to near the noise floor. $\mu = 0.1$ diverges.

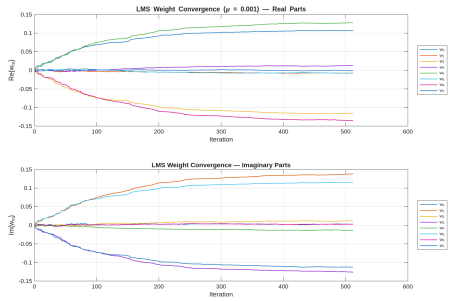


(b) SINR vs iteration. Stable step sizes converge above the Phase 1 fixed-weight baseline (dashed).

Figure 6: Phase 2.1 — LMS convergence behaviour.



(a) Radiation pattern: Phase 1 fixed (blue) vs LMS converged (red dashed) vs uniform initial (green dotted). LMS closely matches the Phase 1 pattern.



(b) All 8 weight trajectories (real and imaginary parts) from zero to steady state over 512 iterations.

Figure 7: Phase 2.1 — LMS converged pattern and weight trajectories.

Phase 2.2 — RLS Adaptive Beamformer Simulation

The RLS adaptive beamformer was simulated using the same signal environment as Phases 1.1 and 2.1. The RLS update minimises the exponentially weighted sum of all past squared errors:

$$\mathbf{k}(n) = \frac{\mathbf{P}(n-1)\,\mathbf{x}(n)}{\lambda + \mathbf{x}^H(n)\,\mathbf{P}(n-1)\,\mathbf{x}(n)}, \quad \mathbf{w}(n) = \mathbf{w}(n-1) + \mathbf{k}(n)\,e^*(n), \quad \mathbf{P}(n) = \frac{1}{\lambda}\left[\mathbf{P}(n-1) - \mathbf{k}(n)\,\mathbf{x}^H(n)\,\mathbf{P}(n-1\right.$$

Forgetting Factor Sweep

Forgetting factor λ	Converged SINR	Notes
0.900	+17.17 dB	Fast tracking, higher misadjustment
0.950	+18.26 dB	Moderate
0.990	+18.80 dB	Good balance
0.999	+18.90 dB	Near-optimal
1.000	**+18.90 dB**	**Best — stationary channel**

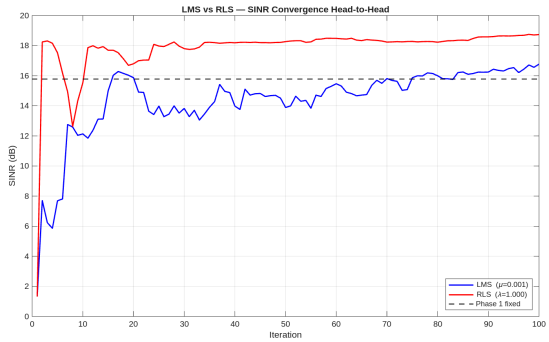
Table 11: Phase 2.2 — RLS forgetting factor sweep results.

Results Summary and Comparison

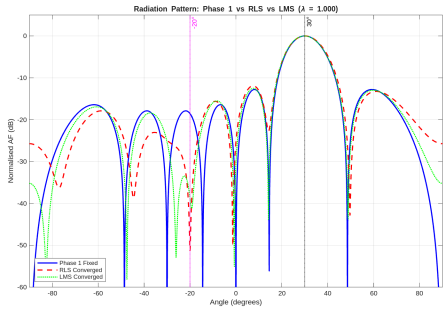
Algorithm	Converged SINR	Convergence sample	Complexity
Phase 1 Fixed	+15.78 dB	N/A (no adaptation)	O(1)
LMS ($\mu = 0.001$)	+18.93 dB	15	O(M)
RLS ($\lambda = 1.000$)	**+18.90 dB**	**2**	O(M^2)

Table 12: Phase 2 — Algorithm comparison summary.

RLS converges in just **2 pilot samples** vs LMS's 15, at a cost of O(M^2) = O(64) operations per sample vs LMS's O(M) = O(8). Both adaptive algorithms exceed the Phase 1 fixed-weight baseline by over 3 dB. For a stationary channel ($\lambda = 1$), RLS is equivalent to a batch least-squares solution — optimal for the available data.



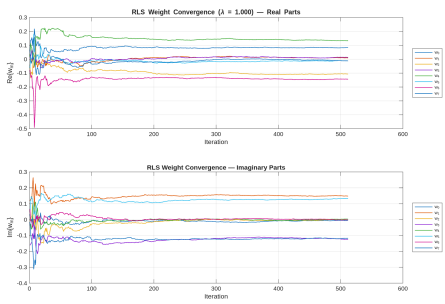
(a) LMS vs RLS SINR convergence (first 100 samples). RLS exceeds the Phase 1 baseline from sample 2; LMS takes ∼80 samples to fully settle.



(b) Radiation patterns: Phase 1 fixed (blue), RLS converged (red dashed), LMS converged (green dotted). All three produce nearly identical beams at 30°.

Figure 8: Phase 2.2 — RLS results and head-to-head comparison.

(a) RLS learning curves. All forgetting factors converge rapidly; $\lambda < 0.95$ shows higher steady-state error due to excessive down-weighting of past data.



(b) RLS weight trajectories settle within the first ∼20 iterations, far faster than LMS.

Figure 9: Phase 2.2 — RLS convergence behaviour.

Phase 2.3 — Algorithm Comparison and Recommendation

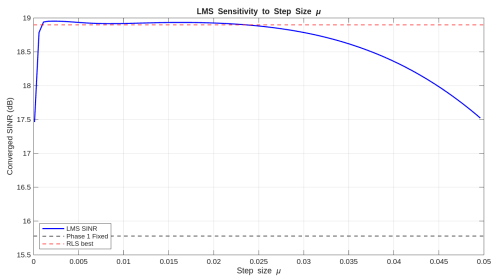
Consolidated Results

Algorithm	SINR	vs Fixed	Conv. sample	MACs/sample
Phase 1 Fixed	+15.78 dB	—	N/A	$O(M) = 8$
LMS (μ=0.001)	+18.93 dB	+3.15 dB	15	$O(M) = 32$
RLS (λ=1.000)	+18.90 dB	+3.12 dB	2	$O(M^2) = 288$

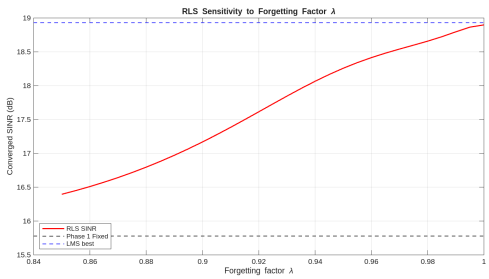
Table 13: Phase 2.3 — Final comparison across all three beamformers.

Sensitivity Analysis

LMS is robust over $\mu \in [0.001, 0.010]$ with SINR remaining above 18.8 dB across the stable range. RLS maintains near-peak SINR for $\lambda \in [0.95, 1.000]$; $\lambda < 0.9$ causes excessive misadjustment. Both algorithms outperform the Phase 1 fixed baseline by over 3 dB across their respective stable parameter ranges.

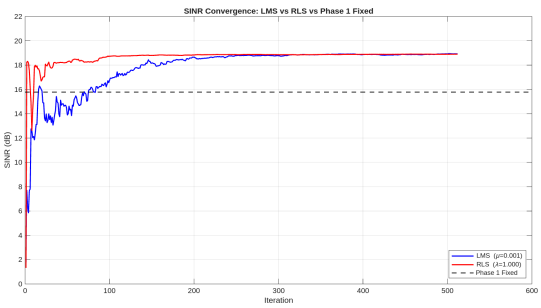


(a) LMS converged SINR vs step size μ. Flat plateau from $\mu = 0.001$ to 0.010; SINR degrades gracefully above $\mu = 0.02$.

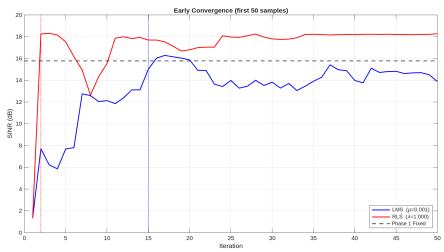


(b) RLS converged SINR vs forgetting factor λ. Near-flat for $\lambda \geq 0.97$; drops sharply below $\lambda = 0.92$.

Figure 10: Phase 2.3 — Sensitivity of LMS and RLS to parameter choice.



(a) Full SINR convergence over 512 samples. RLS (red) leads throughout; LMS (blue) catches up by ∼200 samples. Both settle 3 dB above the Phase 1 dashed baseline.



(b) Zoom on first 50 samples. RLS exceeds the Phase 1 baseline at sample 2; LMS crosses it at sample 15.

Figure 11: Phase 2.3 — SINR convergence comparison.

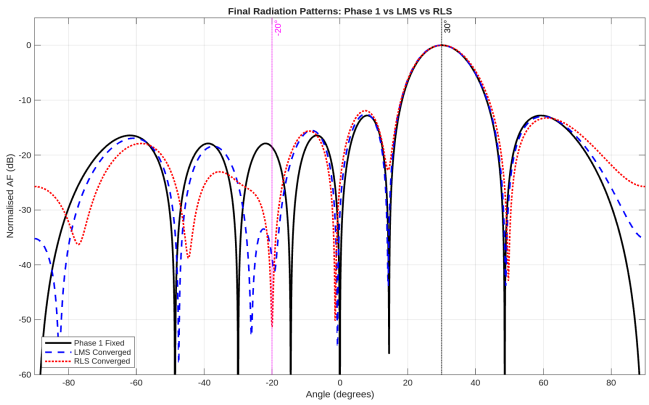


Figure 12: Phase 2.3 — Final radiation patterns. All three beamformers produce a main lobe at 30° with suppression near the interferer at −20. LMS and RLS patterns differ slightly from Phase 1 in sidelobe structure, reflecting the MSE optimisation criterion rather than pure phase steering.

Recommendation for Future Hardware Implementation

For a stationary 5G NR channel with DMRS pilot symbols:

Recommended algorithm: LMS with $\mu = 0.001$–0.005.

- LMS and RLS achieve near-identical steady-state SINR (18.93 dB vs 18.90 dB, $\Delta < 0.03$ dB).
- LMS converges in 15 pilot samples — well within the 5G NR DMRS pilot length of ∼100–200 samples per slot.
- LMS requires 32 MACs/sample vs RLS's 288 MACs/sample (8× lower hardware cost).
- The LMS update maps directly onto the Phase 1 DSP48 pipeline: M parallel multiply-accumulate units, same structure as the existing data path.
- RLS would be preferred for a non-stationary channel (fast-moving user) where $\lambda < 1$ is needed for tracking. In the static lab scenario simulated here, LMS is sufficient.

All phases complete. For full design rationale see `DESIGN.md` v2.0. GitHub: KartikPandeyindia/260410-beamforming.

Generated at end of Phase 1 (Phases 1.1–1.5 complete). Phase 2 direction revised to MATLAB-only algorithmic study — see `DESIGN.md` v2.0 for full rationale. GitHub: KartikPandeyindia/260410-beamforming.